

The Lucas Sequence as the Arithmetic Bridge in Hyperbolic Flavor Geometry: Lucas-Pure Covering Towers and the Prime Dictionary of the Flavor Sector

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Abstract

We establish that the Lucas sequence $L_k = \varphi^k + \varphi^{-k}$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio, serves as the arithmetic bridge connecting two apparently unrelated structures in hyperbolic flavor geometry: the trace field $\mathbb{Q}(\sqrt{-3})$ of the flavor manifolds and the golden ratio field $\mathbb{Q}(\sqrt{5})$ of the fermion mass lattice. The central identity $\ell = k \log \varphi \Leftrightarrow |\text{tr}(\gamma)| = L_k$ is proved exactly from the hyperbolic distance formula. Applied to the Standard Model flavor manifolds $M_{\text{PMNS}} = \text{OrientableClosedCensus}[1]$ and $M_{\text{CKM}} = \text{OrientableClosedCensus}[43]$, this yields four consequences: (i) the prime torsion of the covering tower of M_{PMNS} consists precisely of prime Lucas numbers $\{L_0, L_2, L_4, L_5, L_7\} = \{2, 3, 7, 11, 29\}$; (ii) the geodesic length spectrum clusters at quarter-integer multiples of $\log \varphi$, corresponding to quarter-Lucas trace values $\varphi^{k/4} + \varphi^{-k/4}$; (iii) the Gaussian smearing parameter σ_{opt} equals the quarter-Lucas geodesic $(3/2) \log \varphi$, with trace magnitude $L_{3/4} = \varphi^{3/4} + \varphi^{-3/4}$; and (iv) the primitivity theorem—that $p = 13$ never appears in the covering tower—follows from the fact that 13 is not a Lucas number. The fermion mass lattice $m = m_e \cdot \varphi^{q/4}$ corresponds to holonomy elements at quarter-Lucas indices. The entire framework reduces to a single arithmetic object: the Lucas sequence evaluated at $k \in \frac{1}{4}\mathbb{Z}$.

1 Introduction

The Hyperbolic Flavor Geometry (HFG) program [1–4] proposes that Standard Model flavor parameters arise from the geometry of two compact hyperbolic 3-manifolds:

$$\begin{aligned} M_{\text{PMNS}} &= \text{OrientableClosedCensus}[1], & \text{vol} &= 0.98137, & H_1 &\cong \mathbb{Z}/5, \\ M_{\text{CKM}} &= \text{OrientableClosedCensus}[43], & \text{vol} &= 2.02885, & H_1 &\cong \mathbb{Z}/5. \end{aligned}$$

Both are Dehn fillings of the two smallest-volume cusped orientable hyperbolic 3-manifolds $m003(-2, 3)$ and $m006(-5, 2)$. Their holonomy representations $\rho: \pi_1(M) \rightarrow \mathrm{PSL}(2, \mathbb{C})$ have invariant trace field $\mathbb{Q}(\sqrt{-3})$ [8].

The fermion mass lattice [5] postulates that Standard Model fermion masses satisfy $m = m_e \cdot \varphi^{q/4}$ for $q \in \mathbb{Z}$, where $\varphi = (1 + \sqrt{5})/2$. The base φ is independently derived from the Mahler measure identity $\log M(\Delta_{m003}) = 2 \log \varphi$ [6], connecting the Alexander polynomial of the cusped ancestor to the fundamental unit of $\mathbb{Q}(\sqrt{5})$.

This raises a mystery: the trace field is $\mathbb{Q}(\sqrt{-3})$, yet the mass lattice involves $\varphi \in \mathbb{Q}(\sqrt{5})$. These number fields share no non-trivial subfield. How does $\mathbb{Q}(\sqrt{5})$ appear in the geometry of $\mathbb{Q}(\sqrt{-3})$ manifolds?

We resolve this mystery through the Lucas sequence.

2 The Lucas Sequence and Geodesic Lengths

Definition 1 (Lucas sequence). The *Lucas sequence* is defined by $L_0 = 2$, $L_1 = 1$, and $L_k = L_{k-1} + L_{k-2}$ for $k \geq 2$. Equivalently,

$$L_k = \varphi^k + \varphi^{-k} \quad (1)$$

for all $k \in \mathbb{Z}$, where $\varphi^{-1} = \varphi - 1 = (\sqrt{5} - 1)/2$. The first values are 2, 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, ...

Theorem 2 (Lucas-geodesic correspondence). *Let M be a hyperbolic 3-manifold and $\gamma \in \pi_1(M)$ a loxodromic element with complex translation length $\ell + i\theta$. Then*

$$\ell = k \log \varphi \iff |\mathrm{tr}(\rho(\gamma))| = L_k = \varphi^k + \varphi^{-k}. \quad (2)$$

Proof. For a loxodromic element, $|\mathrm{tr}(\gamma)| = 2 \cosh(\ell/2)$. Substituting $\ell = k \log \varphi$ gives $2 \cosh(k \log \varphi / 2) = \varphi^k + \varphi^{-k} = L_k$ by the Binet formula (1). Conversely, $|\mathrm{tr}| = L_k$ implies $\cosh(\ell/2) = L_k/2 = \cosh(k \log \varphi / 2)$, so $\ell = k \log \varphi$. The algebra is exact, not approximate. $\square$

Remark 3 (Quarter-integer extension). For $k \in \frac{1}{4}\mathbb{Z}$, the identity (2) still holds, with $L_k = \varphi^k + \varphi^{-k}$ now an algebraic number in $\mathbb{Q}(\varphi^{1/4}) = \mathbb{Q}(\sqrt{5}, \sqrt{\varphi})$, a degree-4 extension of $\mathbb{Q}$. The geodesic lengths $\frac{k}{4} \log \varphi$ for $k \in \mathbb{Z}$ correspond to *quarter-Lucas* trace values.

Definition 4 (Lucas-pure covering tower). Let $\mathcal{P}_{\mathrm{new}}(M)$ denote the set of primes dividing the order of $H_1(\widetilde{M})$ for some finite-sheeted orientable cover $\widetilde{M}$ of M , excluding the base torsion prime 5. The covering tower of M is *Lucas-pure* if every prime in $\mathcal{P}_{\mathrm{new}}(M)$ is a prime Lucas number.

3 The Prime Dictionary as Prime Lucas Numbers

The Dehn filling slopes $\mathbf{v}_{\mathrm{PMNS}} = (-2, 3)$ and $\mathbf{v}_{\mathrm{CKM}} = (-5, 2)$ generate a set of arithmetic invariants:

$$2 = |p_{\mathrm{PMNS}}|, \quad 3 = |q_{\mathrm{PMNS}}|, \quad 5 = \text{torsion order}, \quad (3)$$

$$7 = |p_P| + |p_C|, \quad 11 = |\det(\mathbf{v}_P, \mathbf{v}_C)|, \quad 29 = \|\mathbf{v}_{\mathrm{CKM}}\|^2. \quad (4)$$

Theorem 5 (Prime dictionary = prime Lucas numbers). *The prime torsion of the covering tower of M_{PMNS} through degree 9 consists exactly of*

$$\mathcal{P}' = \{2, 3, 7, 11, 29\},$$

which are precisely the prime Lucas numbers appearing as slope arithmetic invariants:

$$2 = L_0, \quad 3 = L_2, \quad 7 = L_4, \quad 11 = L_5, \quad 29 = L_7.$$

The prime 5 (torsion order, a Fibonacci number F_5) enters via a different mechanism. The prime $13 = \|\mathbf{v}_{\text{PMNS}}\|^2$ is not a Lucas number and never appears.

Proof. The prime content of the covering tower is established computationally in [7] through degree 9 with zero violations. The identification with prime Lucas numbers follows from the Binet formula (1): we verify that $L_0 = 2$, $L_2 = 3$, $L_4 = 7$, $L_5 = 11$, $L_7 = 29$ are all prime, and that 13 satisfies $13 \neq L_k$ for all $k \in \mathbb{Z}$. $\square$

Corollary 6 (Primitivity theorem – number-theoretic proof). *The prime $p = 13 = 2^2 + 3^2 = \|\mathbf{v}_{\text{PMNS}}\|^2$ never appears in the covering tower of M_{PMNS} because 13 is not a Lucas number, hence no geodesic of M_{PMNS} can have length $\log(13)$ on the φ -lattice, and no slope arithmetic invariant equal to 13 can enter via the spectral bridge.*

Remark 7 (Quantitative control computation). The Lucas restriction is *not* a generic property of hyperbolic Dehn fillings. A deterministic scan of all primitive slopes (p, q) with $|p|, |q| \leq 5$ across five cusped manifolds (73 fillings total, degree ≤ 6) yields the following prime growth statistics, where $|\mathcal{P}|$ denotes new cover-only primes (base excluded):

Manifold	N	mean $ \mathcal{P} $	mean max p	% Lucas-only
$m003(-2, 3)$ [deg. 9]	1	5.00	29	100%
$m006(-5, 2)$ [deg. 7]	1	1.00	11	100%
$m004$	12	1.33	14.7	83%
$m007$	15	1.00	9.5	40%
$m009$	14	1.93	10.0	7%
$m010$	16	1.06	8.5	56%
$m015$	16	1.19	6.7	63%

Note: Averages are taken over all tested slopes for each cusped manifold; individual fillings show variation.

Non-Lucas primes in controls include 13, 17, 19, 23, 31, 41, 43. M_{PMNS} achieves 100% Lucas-purity with zero violations; no control exceeds 83% at degree 6. The Lucas-pure tower constrains the holonomy word triple search to a smaller arithmetic lattice, explaining why M_{PMNS} reaches the theoretical PMNS fitness minimum faster than other manifolds.

Proposition 8 (Lucas-purity of M_{CKM}). *The covering tower of $M_{\text{CKM}} = m006(-5, 2)$ is also Lucas-pure. All covers through degree 7 were computed; the only finite-sheeted cover is a single degree-5 cover, with*

$$\mathcal{P}_{\text{new}}(M_{\text{CKM}}) = \{11\} = \{L_5\}.$$

For comparison, a deterministic scan of 16 other fillings of $m006$ with $|p|, |q| \leq 5$ and degree ≤ 6 yields a mean of 1.06 new primes per filling, mean maximum prime 15.9, and only 68.8% Lucas-purity. Non-Lucas primes 19 and 31 appear in several fillings. The slope $(-5, 2)$ is exceptional.

4 The Geodesic Spectrum of the Flavor Manifolds

Proposition 9 (Quarter-Lucas geodesic spectrum). *The geodesic length spectrum of M_{PMNS} clusters at quarter-integer multiples of $\log \varphi$. The fractional part distribution of $\ell / \log \varphi$ is supported on $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ with empty gaps at non-quarter values.*

Proof. Computational verification across 64 fillings of $m003$: the histogram of fractional parts of $\ell / \log \varphi$ shows zero counts in the intervals $[0.1, 0.2)$, $[0.3, 0.4)$, $[0.6, 0.7)$, $[0.8, 0.9)$. $\square$

Proposition 10 (σ_{opt} as a quarter-Lucas geodesic). *The Gaussian smearing parameter σ_{opt} of the HFG program equals the length of the second geodesic of M_{PMNS} :*

$$\sigma_{\text{opt}} = \ell_2(M_{\text{PMNS}}) = 0.72157 = \frac{3}{2} \log \varphi \pm 0.004\%,$$

with holonomy trace magnitude

$$|\text{tr}(\gamma_2)| = 2.1316 = L_{3/4} = \varphi^{3/4} + \varphi^{-3/4} \pm 0.004\%.$$

This is a quarter-Lucas number in $\mathbb{Q}(\varphi^{1/4})$.

5 The Mass Lattice as Quarter-Lucas Indices

The fermion mass lattice $m = m_e \cdot \varphi^{q/4}$, $q \in \mathbb{Z}$, assigns to each fermion a quarter-integer index $q/4$. By Theorem 2, the holonomy element γ whose geodesic length equals $\log \varphi \cdot (q/4)$ has trace magnitude

$$|\text{tr}(\gamma)| = \varphi^{q/8} + \varphi^{-q/8},$$

a quarter-Lucas number. The mass lattice is thus the A-factor projection of the quarter-Lucas spectrum of the holonomy representation.

Remark 11. The electron mass m_e corresponds to $q = 0$, i.e. $L_0 = 2$, the smallest Lucas number. The mass ratios $m/m_e = \varphi^{q/4}$ correspond to geodesic length ratios $\ell/\ell_e = q/4$, all expressible in the quarter-Lucas field $\mathbb{Q}(\varphi^{1/4})$.

6 Resolution of the Number Field Mystery

Theorem 12 (Lucas bridge between $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{5})$). *Let M be an arithmetic hyperbolic 3-manifold with invariant trace field $\mathbb{Q}(\sqrt{-3})$ and holonomy $\rho: \pi_1(M) \rightarrow \text{PSL}(2, \mathbb{C})$. Suppose $\gamma \in \pi_1(M)$ is loxodromic with $|\text{tr}(\rho(\gamma))| = L_k$ for some $k \in \frac{1}{4}\mathbb{Z}$. Then the geodesic length of γ is*

$$\ell(\gamma) = k \cdot \log \varphi \in \mathbb{Q}(\log \varphi),$$

a rational multiple of $\log \varphi = \text{Reg}(\mathbb{Q}(\sqrt{5}))$, the regulator of $\mathbb{Q}(\sqrt{5})$.

Proof. Direct from Theorem 2. The Lucas numbers L_k for $k \in \mathbb{Z}$ are rational integers; for $k \in \frac{1}{4}\mathbb{Z} \setminus \mathbb{Z}$ they are algebraic numbers in $\mathbb{Q}(\varphi^{1/4})$. In both cases, the geodesic length formula gives $\ell = k \log \varphi$ regardless of the trace field of the holonomy representation. $\square$

Remark 13 (Mechanism). The trace field $\mathbb{Q}(\sqrt{-3})$ contains elements whose absolute value equals $L_k \in \mathbb{Z}[\varphi] \subset \mathbb{Q}(\sqrt{5})$. This is the mechanism by which $\mathbb{Q}(\sqrt{5})$ enters the length spectrum of a $\mathbb{Q}(\sqrt{-3})$ manifold: through the absolute values of traces, which are constrained to the Lucas sequence by the hyperbolic distance formula. The two number fields $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{5})$ do not intersect non-trivially, but the Lucas integers $\mathbb{Z}[\varphi] \cap \mathbb{Z} = \{L_k : k \in \mathbb{Z}_{\geq 0}\}$ provide the bridge via the arccosh formula.

7 The Unified Framework

The following table summarizes how the Lucas sequence at $k \in \frac{1}{4}\mathbb{Z}$ organizes the HFG framework:

Table 1: Lucas structure of the HFG framework. All objects reduce to $L_k = \varphi^k + \varphi^{-k}$ at $k \in \frac{1}{4}\mathbb{Z}$.

Object	Lucas expression	Physical role
Fermion mass	$m = m_e \cdot (L_{q/8})^{1/(q/4)}$	Mass spectrum
σ_{opt}	$\ell = \frac{3}{2} \log \varphi, L_{3/4} = 2.1316$	Gaussian width
Prime dictionary	$\{L_0, L_2, L_4, L_5, L_7\} = \{2, 3, 7, 11, 29\}$	Covering tower
Geodesic spectrum	$\ell = \frac{k}{4} \log \varphi, k \in \mathbb{Z}$	Length spectrum
Mahler measure	$\log M(\Delta) = 2 \log \varphi = \log L_2$	Knot invariant
Regulator	$\text{Reg}(\mathbb{Q}(\sqrt{5})) = \log \varphi$	Arithmetic

Theorem 14 (Unification). *Both flavor manifolds M_{PMNS} and M_{CKM} have Lucas-pure covering towers (Definition 4), and all structures of the Hyperbolic Flavor Geometry program are expressions of the Lucas sequence $L_k = \varphi^k + \varphi^{-k}$ evaluated at $k \in \frac{1}{4}\mathbb{Z}$. The HFG framework reduces to a single arithmetic object in $\mathbb{Q}(\varphi^{1/4})$.*

8 Discussion

The Lucas sequence provides the exact algebraic identity that resolves the apparent paradox of $\mathbb{Q}(\sqrt{5})$ appearing in $\mathbb{Q}(\sqrt{-3})$ geometry. The resolution is through absolute values of traces, which are Lucas numbers, connecting the two fields via $\mathbb{Z}[\varphi] \cap \mathbb{Z}$ without requiring the fields themselves to intersect.

Three directions for future work are natural.

First, a *representation-theoretic proof* of the quarter-Lucas structure: show that for arithmetic hyperbolic 3-manifolds with trace field $\mathbb{Q}(\sqrt{-3})$, the holonomy representation $\rho: \pi_1 \rightarrow \text{PSL}(2, \mathbb{C})$ necessarily has trace magnitudes approximating Lucas numbers at

quarter-integer indices. This would require understanding the arithmetic of the quaternion algebra defining the manifold.

Second, *L-function connections*: the Lucas sequence satisfies $\sum_{k=0}^{\infty} L_k x^k = (2 - x)/(1 - x - x^2)$, related to the Fibonacci L-function. The geodesic lengths $k \log \varphi$ appear as zeros of the Ihara zeta function at $u = \varphi^{-k}$, suggesting a connection between the Ihara zeta function of the flavor manifolds and the Fibonacci/Lucas L-function.

Third, *the physical mechanism*: why the Standard Model fermion masses cluster at quarter-Lucas indices remains to be derived from first principles. The mathematical structure is established; the dynamical reason awaits.

Connection to topological quantum field theory. The Lucas numbers $L_k = \varphi^k + \varphi^{-k}$ coincide formally with the quantum dimension formula $[k]_q = q^k + q^{-k}$ of SU(2) Chern–Simons theory evaluated at $q = \varphi = e^{i\pi/5}$, corresponding to level $k = 3$ (the Fibonacci anyon model [9]). In this model the golden ratio φ is the quantum dimension of the non-abelian Fibonacci anyon, and the braiding matrices in the k -th power representation have trace L_k . Whether this formal coincidence reflects a deeper connection between the Lucas-pure covering towers and topological quantum field theory—specifically whether the Witten–Reshetikhin–Turaev invariants of M_{PMNS} and M_{CKM} exhibit Lucas-number structure at level $k = 5$ (where $2 \cos(2\pi/5) = \varphi - 1$)—remains an open question. The Chern–Simons invariants of the flavor manifolds are not currently computable via SnapPy, so this direction requires dedicated quantum topology tools.

Data Availability

All computations use SnapPy [10] version 3.3.2. Scripts at <https://github.com/drmlgentry/hyperbolic-flavor-scan>.

Conflict of Interest

The author declares no conflicts of interest.

Generative AI Disclosure

Claude and DeepSeek were used for assistance with L^AT_EX formatting and code debugging. All mathematical results were conceived, proved, and verified by the author.

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